Probability and Stochastic Processes

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Chapter 5 Introduction to Stochastic Processes

Probability provides models for analyzing random or unpredictable outcomes. The main new ingredient in stochastic processes is the explicit role of time. A stochastic process is described by its position $X(t)$ at time $t \in T \subseteq [0, \infty)$. T may be $[0, 1]$, $[0, \infty)$, $\{0, 1, 2, \cdots\}$ and so on. The theory of stochastic processes is concerned with the investigation of the structure of families of random variables $X(t)$. The random variable could be the location of a particle on the real line, the strength of a signal, the price of a stock, and many other possibilities as well.

In this chapter, the objects we deal with are countable infinite and uncountable infinite random variables.

Contents

1. Definition and Classification
2. The Distribution Family and the Moment Functions
3. The Moments of the Stochastic Processes
4. Stochastic Analysis

Introduction to Stochastic Processes

A random variable X is a rule for assigning to every outcome ω of an experiment E a number $X(\omega)$. A stochastic process $X(t)$ is a rule for assigning to every ω a function $X(t, \omega)$. Thus a ***stochastic process*** is ***a family of time functions*** depending on the parameter ω. Each time function is called a sample path. On the other hand, a stochastic process $\{X(t), t \in T\}$ is ***a family of random variables***. That is, for a given $t \in T$, $X(t, \omega)$ is a function of sample point ω, i.e., a random variable. Or, equivalently, a stochastic process $X(t)$ is a function of t and ω. The domain of ω is the sample space Ω and the domain T of t is a subset of real numbers.

Introduction to Stochastic Processes

The complete description of a stochastic is $\{X(t,\omega), t \in T\}$, but we usually use the abbreviation $X(t)$ or X_t.

- T the index set of the process
- t often interpreted as time

As a result, we refer to X_t as the state of the process at time t.

For example, X_t might equal the total number of customers that have entered a supermarket by time t; or the number of customers in the supermarket at time t; or the total amount of sales that have been recorded in the market by time t; etc. The state space of a stochastic process is defined as the set of all possible values that the random variables X_t can assume.

Introduction to Stochastic Processes

For example, consider the experiment consisting of a coin toss. A suitable sample space would be $\Omega = \{H, T\}$. Let

$$X_n(\omega) = \begin{cases} 0 & \text{if the outcome of } n\text{th is } \omega = T, \\ 1 & \text{if the outcome of } n\text{th is } \omega = H. \end{cases}$$

where $n = 1, 2, \cdots$. If we do this experiment 5 times, then we can get the following results

$$H, H, T, T, H$$

Observe the outcome $\omega = H$ now. Then we get

$$X_1 = 1,\ X_2 = 1,\ X_3 = 0,\ X_4 = 0,\ X_5 = 1.$$

Introduction to Stochastic Processes

We can plot the sequence, a sample path. Since we do the experiment 5 times, we can get 2^5 sample paths. As you can imagine, we would get a family of sample paths if we do infinite times.

On the other hand, for a given index n, X_n is a random variable. Since $n = 1, 2, \cdots$, we then obtain a family of random variables.

Example

In the coin-tossing experiment we define X_t such that

$$X_t = \begin{cases} \sin t & \text{for } \omega = H, \\ 2t & \text{for } \omega = T. \end{cases}$$

Thus X_t consists of two very regular curves. It is, nevertheless, a stochastic process.

Example

Consider

$$X_t = acos(\omega t + \Theta),\ t \in (-\infty, +\infty),$$

where a, ω are constants and Θ has a uniform distribution in the interval $(0, 2\pi)$. Since for each fixed $t = t_1$, $X_{t_1} = acos(\omega t_1 + \Theta)$ is a random variable, $\{X_t\}$ is a stochastic process. If we select $\theta_i \in (0, 2\pi)$ arbitrarily, then

$$X_t = acos(\omega t + \theta_i),\ t \in (-\infty, +\infty),$$

is a sample path of the process. $\square$

Introduction to Stochastic Processes

The main elements distinguishing stochastic processes are in the nature of the state space S, the index parameter T, and the dependence relations among the random variables X_t.

State Space S
This is the space in which the possible values of each X_t lie. In the case that $S = \{0, 1, 2, \cdots\}$, we refer to the process at hand as integer valued, or alternately as a discrete state process. If $S = (-\infty, \infty)$, then we call X_t a real-valued stochastic process. If S is Euclidean space of dimension k, then X_t is said to be a k-vector process.

As in the case of a single random variable, the choice of state space is not uniquely specified by the physical situation being described, although usually one particular choice stands out as most appropriate.

Index Parameter T

When T is a countable set the stochastic process is said to be a discrete-time process, or a stochastic sequence. Often when T is discrete we shall write X_n instead of X_t. We have discussed the family of distribution functions and the moments of a stochastic sequence in section 4.1. If T is an interval of the real line, then the stochastic process is said to be a continuous-time process. For instance, $\{X_n, n = 0, 1, \cdots\}$ is a discrete-time stochastic process indexed by the nonnegative integers; while $\{X_t, t \geqslant 0\}$ is a continuous-time stochastic process indexed by the nonnegative real numbers.

The index set T can be not one dimensional. One example is that of waves in oceans. We may regard the latitude and longitude coordinates as the value of t, and X_t is then the height of the wave at the location t.

Classical Types of Stochastic Processes

We now describe some of the classical types of stochastic processes characterized by different dependence relationships among X_t. In the examples, we take $T = [0, \infty)$ unless we state the contrary explicitly. For simplicity of exposition, we assume that the random variables X_t are real valued.

(a) ***Markov Processes***

Roughly speaking, a Markov process is a process with the property that, given the value of X_t, the values of $X_s, s > t$, do not depend on the values of $X_u, u < t$; that is, the probability of any particular future behavior of the process, when its present state is known exactly, is not altered by additional knowledge concerning its past behavior.

We should make it clear, however, that if our knowledge of the present state of the process is imprecise, then the probability of some future behavior will in general be altered by additional information relating to the past behavior of the system.

Introduction to Stochastic Processes

In formal terms a process is said to be Markovian if

$$\begin{aligned} &P(a < X_t \leqslant b \,|X_{t_1} = x_1, X_{t_2} = x_2, \cdots X_{t_n} = x_n) \\ =&P(a < X_t \leqslant b \,|X_{t_n} = x_n) \end{aligned}$$

whenever $t_1 < t_2 < \cdots < t_n < t$.

Markov chains, as statistical models of real-world processes, have many applications in a wide range of topics such as physics, chemistry, medicine, music, game theory and sports. It will be introduced in next chapter.

(b) ***Stationary Processes***

A stochastic process X_t for $t \in T$ is said to be *strictly stationary* if the joint distribution functions of the families of random variables

$$(X_{t_1+h}, X_{t_2+h}, \cdots, X_{t_n+h}) \quad \text{and} \quad (X_{t_1}, X_{t_2}, \cdots, X_{t_n})$$

are the same for all $h > 0$ and arbitrary selections $t_1, t_2, \cdots, t_n$ from T. This condition asserts that in essence the process is in probabilistic equilibrium and that the particular times at which we examine the process are of no relevance. In particular, the distribution of X_t is the same for each t.

Introduction to Stochastic Processes

A stochastic process X_t for $t \in T$ is said to be *wide sense stationary* or *covariance stationary* if it possesses finite second moments and if

$$Cov(X_t, X_{t+h}) = E(X_t X_{t+h}) - E(X_t)E(X_{t+h})$$

depends only on h for all $t \in T$. A strict stationary process that has finite second moments is wide stationary. Stationary

processes are appropriate for describing many phenomena that occur in communication theory, astronomy, biology, and sometimes economics and are discussed in more detail in Chapter 7.

(c) ***Process with Stationary Independent Increments***

We say that X_t is *a process with independent increments* if for any $0 \leqslant t_0 < t_1 < t_2 < \cdots < t_n$,

$$X_{t_1} - X_{t_0}, X_{t_2} - X_{t_1}, \cdots, X_{t_n} - X_{t_{n-1}}$$

are independent. If the index set is discrete, that is, $T = (0, 1, \cdots)$, then a process with independent increments reduces to a sequence of independent random variables $Z_0 = X_0, Z_i = X_i - X_{i-1} (i = 1, 2, 3, \cdots)$, in the sense that knowing the individual distributions of $Z_0, Z_1, \cdots$. enables one to determine (as should be fairly clear to the reader) the joint distribution of any finite set of the X_i.

In fact,

$$X_i = Z_0 + Z_1 + \cdots + Z_i, \quad i = 0, 1, 2, \cdots$$

If the distribution of the increments $X_{t_1+h} - X_{t_1}$ depends only on the length h of the interval and not on the time t_1, then the process is said to have *stationary increments*. For a process with stationary increments the distribution of $X_{t_1+h} - X_{t_1}$ is the same as the distribution of $X_{t_2+h} - X_{t_2}$, no matter what the values of t_1, t_2 and h.

We will put our attention to two independent increments processes, Poisson process and Wiener process in Chapter 8, which are useful in modeling a large number of random phenomena.

Contents

1. Definition and Classification
2. The Distribution Family and the Moment Functions
3. The Moments of the Stochastic Processes
4. Stochastic Analysis

The Distribution Family and the Moment Functions

Similar to random sequences, we can define the family of distribution functions of a stochastic process which has an uncountable infinity of random variables, one for each t. For a specific t, X_t is a random variable with distribution

$$F(x;t) = P\{X_t \leqslant x\}.$$

The distribution function of this random variable, in general, depend on t. The function $F(x;t)$ is called the first-order distribution function of the process X_t. Its derivative with respect to x:

$$f(x,t) = \frac{\partial F(x,t)}{\partial x}$$

is the first-order density of X_t.

The Distribution Family and the Moment Functions

The second-order distribution of the process X_t is the joint distribution

$$F(x_1, x_2; t_1, t_2) = P\big(X_{t_1} \leq x_1, X_{t_2} \leq x_2\big).$$

The corresponding density equals

$$f(x_1, x_2; t_1, t_2) = \frac{\partial^2 F(x_1, x_2, ; t_1, t_2)}{\partial x_1 \partial x_2}.$$

The Distribution Family and the Moment Functions

Similar to random sequence, if we want to determine the whole statistical property of a stochastic process X_t, then we need the information of the ***family of distribution functions*** of X_t,

$$\{F(x_1, x_2, \ldots, x_n; t_1, t_2, \ldots, t_n), \forall n \geqslant 1,\ \forall t_i \in T, i = 1, 2, \ldots, n\}.$$

That is, we should master the probability distribution of this vector for all times t_1 through t_n and for all positive integers n.

The Distribution Family and the Moment Functions

Sometimes we need consider two stochastic processes at the same time. For example, when considering a wide-sense stationary processes through linear time-invariant systems, it is conventional to consider both input and output processes.

Definition

Suppose that X_t and Y_t are two stochastic processes defined on the same sample space Ω and they have the same parameter set T. If for all $t \in T$, (X_t, Y_t) is a random variable, then we say that $\{(X_t, Y_t), t \in T\}$ is a two-dimensional process.

The Distribution Family and the Moment Functions

Definition

Suppose that $\{(X_t, Y_t), t \in T\}$ *is a two-dimensional process. For all positive integers* n *and* m*, and for all* $t_1, t_2, \ldots, t_n$*;* $t'_1, t'_2, \ldots, t'_m \in T$*, the* $(n+m)$*th-order probability distribution function of the process* (X_t, Y_t) *is defined by*

$$\begin{aligned} & F(x_1, \ldots, x_n; y_1, \ldots, y_m; t_1, \ldots, t_n; t'_1, \ldots, t'_m) \\ = & P\{X(t_1) \leqslant x_1, \ldots, X(t_n) \leqslant x_n; Y(t'_1) \leqslant y_1, \ldots, Y(t'_m) \leqslant y_m\}. \end{aligned}$$

The Distribution Family and the Moment Functions

The processes $\{X_t\}$ and $\{Y_t\}$ are independent if the group $\{X(t_1), \cdots, X(t_n)\}$ is independent of the group $\{Y(t'_1), \cdots, Y(t'_m)\}$ for any $n, m \in \mathbb{N}$, $t_1, t_2, \ldots, t_n$, $t'_1, t'_2, \ldots, t'_m \in T$. That is

$$\begin{aligned} & F(x_1, \ldots, x_n; y_1, \ldots, y_m; t_1, \ldots, t_n; t'_1, \ldots, t'_m) \\ = & F_X(x_1, \ldots, x_n; t_1, \ldots, t_n) \cdot F_Y(y_1, \ldots, y_m; t'_1, \ldots, t'_m). \end{aligned}$$

The Distribution Family and the Moment Functions

Definition

Suppose that $\{X_t\}$ *and* $\{Y_t\}$ *are two real-valued processes, we say that*

$$Z_t = X_t + iY_t$$

is a complex process. A complex process is a family of complex functions.

The complex process $Z_t = X_t + iY_t$ is specified in terms of the joint statistics of the real processes X_t and Y_t.
A vector process (n-dimensional process) is a family of n stochastic processes.

Contents

1. Definition and Classification
2. The Distribution Family and the Moment Functions
3. The Moments of the Stochastic Processes
4. Stochastic Analysis

The Moments of the Stochastic Processes

For the determination of the statistical properties of a stochastic process, knowledge of the family of distribution functions of the stochastic process is required for every x_i, t_i and n. However, for many applications, only certain averages are used, in particular, the expected value of X_t and of X_t^2.

The Moments of the Stochastic Processes

Mean, Autocorrelation and Autocovariance

If X_t is a real stochastic process, then its mean function is

$$\mu_X(t) = E(X_t),\ t \in T.$$

The variance function is

$$\sigma_X^2(t) = Var(X_t).$$

The autocorrelation function is

$$R_X(t_1, t_2) = R_{XX}(t_1, t_2) = E(X_{t_1} X_{t_2}).$$

Correlation functions have special properties. First,

$$R_X(t_1, t_2) = E(X_{t_1} X_{t_2}) = E(X_{t_2} X_{t_1}) = R_X(t_2, t_1).$$

The Moments of the Stochastic Processes

In other words, the correlation function is a symmetric function of t_1 and t_2. Next, observe that $R_X(t,t) = E(X_t^2) \geqslant 0$ and we call it the power function. Then we have for any t_1 and t_2,

$$R_X(t_1, t_2) \leqslant \sqrt{E(X_{t_1}^2)}\sqrt{E(X_{t_2}^2)}.$$

Correlation functions have special properties. First,

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The Moments of the Stochastic Processes

The autocovariance function is

$$\begin{aligned} C_X(t_1, t_2) &:= Cov(X_{t_1}, X_{t_2}) \\ &= E\left[\left(X_{t_1} - \mu_X(t_1)\right)\left(X_{t_2} - \mu_X(t_2)\right)\right]. \end{aligned}$$

Notice that the covariance function is also symmetric, i.e., $C_X(t_1, t_2) = C_X(t_2, t_1)$. By easy calculations,

$$\sigma^2_{X_t} = R_X(t, t) - \mu^2_X(t),$$

$$C_X(t_1, t_2) = R_X(t_1, t_2) - \mu_X(t_1)\mu_X(t_2).$$

The Moments of the Stochastic Processes

Example

An extreme example of a stochastic process is a deterministic signal $X_t = f(t)$. In this case,

$$\mu_X(t) = E(f(t)) = f(t), \;\; R_X(t_1, t_2) = E\big(f(t_1)f(t_2)\big) = f(t_1)f(t_2).$$

Example

Consider the stochastic process

$$X_t = Y cos\omega t + Z sin\omega t, \quad t \geqslant 0,$$

where Y and Z are independent, real-valued random variables and each has a normal distribution with mean 0 and variance σ^2, ω is a constant. Find the mean function $\mu_X(t)$ and autocorrelation function $R_X(s, t)$.

The Moments of the Stochastic Processes

Solution. First

$$\mu_X(t) = E(X_t) = E(Y cos\omega t + Z sin\omega t) = E(Y)cos\omega t + E(Z)sin\omega t =$$

Then, since Y and Z are independent, for the autocorrelation,

$$\begin{aligned} R_X(s,t) &= E(X_s X_t) \\ &= E[(Y cos\omega s + Z sin\omega s)(Y cos\omega t + Z sin\omega t)] \\ &= cos\omega s \cdot cos\omega t \cdot E(Y^2) + sin\omega s \cdot sin\omega t \cdot E(Z^2) \\ &= \sigma^2 cos\omega(t-s). \end{aligned}$$

$\square$

The Moments of the Stochastic Processes

Example

Consider the stochastic process

$$X_t = a\sin(\omega t + \Theta), \quad t \in (-\infty, +\infty),$$

where a and ω are constants and Θ has a uniform distribution in the interval $(0, 2\pi)$. This kind of X_t is called a sine wave with random phase. Find the mean function $\mu_X(t)$, the autocorrelation function $R_X(s,t)$ and the variance function $\sigma_X^2(t)$.

The Moments of the Stochastic Processes

Solution. Since the probability density function of Θ is

$$f(\theta) = \begin{cases} \dfrac{1}{2\pi} & \text{for } 0 < \theta < 2\pi, \\ 0 & \text{otherwise,} \end{cases}$$

we have

$$\mu_X(t) = E[a\sin(\omega t + \Theta)] = \frac{a}{2\pi}\int_0^{2\pi} \sin(\omega t + \theta)d\theta = 0,$$

$$\begin{aligned} R_X(t_1, t_2) &= E[a\sin(\omega t_1 + \Theta) \cdot a\sin(\omega t_2 + \Theta)] \\ &= \frac{a^2}{2\pi}\int_0^{2\pi} \sin(\omega t_1 + \theta)\sin(\omega t_2 + \theta)d\theta \\ &= \frac{a^2}{2} cos(\omega(t_2 - t_1)). \end{aligned}$$

The Moments of the Stochastic Processes

Let $t_2 = t_1 = t$. we get

$$\sigma_X^2(t) = R_X(t,t) - \mu_X^2(t) = \frac{a^2}{2}.$$

$\square$

The Moments of the Stochastic Processes

Example

Suppose that X_t is a process with

$$\mu_X(t) = 3, \qquad R_X(t_1, t_2) = 9 + 4e^{-0.2|t_1 - t_2|}.$$

Determine the mean, the variance, and the covariance of the random variables $Z = X_5$ and $W = X_8$.

Solution. Clearly, $E(Z) = \mu_X(5) = 3$ and $E(W) = \mu_X(8) = 3$. Furthermore,

$$E(Z^2) = R(5,5) = 13, \quad E(W^2) = R(8,8) = 13.$$

$$R_X(5,8) = E(ZW) = R(5,8) = 9 + 4e^{-0.6} = 11.195.$$

Thus Z and W have the same variance $\sigma^2 = 13 - 9 = 4$ and their covariance equals
$C_X(5,8) = R_X(5,8) - E(Z)E(W) = 2.195.$ $\square$

The Moments of the Stochastic Processes

Example

Suppose that the stochastic process X_t is

$$X_t = Y + Zt,$$

where Y and Z are independent and each has a normal distribution $N(0, 1)$.
(a) Find the first-order distribution of $X_{2.5}$.
(b) Find the first-order distribution of X_t.
(c) Find the second-order distributions of X_t.

The Moments of the Stochastic Processes

Solution. (a) Since Y and Z are independent and each has a normal distribution with mean 0 and variance 1, the linear combination of Y and Z

$$X_{2.5} = Y + 2.5Z$$

has a normal distribution with mean 0 and variance 7.25.
(b) For a given $t > 0$, X_t is a linear combination of Y and Z. So X_t has a normal distribution. Since

$$E(X_t) = 0, \quad Var(X_t) = 1 + t^2,$$

X_t has a normal distribution with mean 0 and variance $1 + t^2$.

The Moments of the Stochastic Processes

(c) For two specific s, t and $s \neq t$,

$$\begin{pmatrix} X_s \\ X_t \end{pmatrix} = \begin{pmatrix} 1 & s \\ 1 & t \end{pmatrix} \begin{pmatrix} Y \\ Z \end{pmatrix}.$$

Since Y and Z are independent and each has a normal distribution, (Y, Z) has a bivariate normal distribution. Thus, (X_s, X_t) has a bivariate normal distribution. By an easy calculation,

$$E(X_s) = E(X_t) = 0,\ Var(X_s) = 1 + s^2,\ Var(X_t) = 1 + t^2,$$

$$Cov(X_s, X_t) = E\big((Y + Zs)(Y + Zt)\big) = 1 + st.$$

Thus

$$(X_s, X_t) \sim N\left(0, 0, 1 + s^2, 1 + t^2, \frac{1 + st}{\sqrt{(1 + s^2)(1 + t^2)}}\right).$$

$\square$

The Moments of the Stochastic Processes

In fact, the autocorrelation of a process X_t, real or complex, is by definition the mean of the product $X_{t_1}\overline{X}_{t_2}$. That is

$$R_X(t_1, t_2) = E(X_{t_1}\overline{X}_{t_2}),$$

where the conjugate term associated with the second variable in $R_X(t_1, t_2)$. From this it follows that

$$R_X(t_2, t_1) = E(X_{t_2}\overline{X}_{t_1}) = \overline{R_X(t_1, t_2)}.$$

The autocovariance $C_X(t_1, t_2)$ of a complex process X_t is the covariance of the random variables X_{t_1} and X_{t_2}, which can be defined by

$$C_X(t_1, t_2) = R_X(t_1, t_2) - \mu_X(t_1)\overline{\mu_X(t_2)}.$$

In this book, we merely discuss real case.

The Moments of the Stochastic Processes

Cross-correlation and Cross-covariance

When we observe two stochastic processes, we will concern the cross-corre1ation and cross-covariance of two processes X_t and Y_t.
The cross-correlation function is

$$R_{XY}(t_1, t_2) = E(X_{t_1}\overline{Y}_{t_2}) = \overline{R_{YX}(t_2, t_1)}.$$

Similarly, $C_{XY}(t_1, t_2) = R_{XY}(t_1, t_2) - \mu_X(t_1)\overline{\mu_Y(t_2)}$ is their cross-covariance function.

The Moments of the Stochastic Processes

Since we mainly concern real-valued processes,

$$\overline{Y}_{t_2} = Y_{t_2}, \quad \overline{\mu_Y(t)} = \mu_Y(t), \quad \text{and} \quad R_{XY}(t_1, t_2) = R_{YX}(t_2, t_1).$$

Two processes $\{X_t\}$ and $\{Y_t\}$ are called *uncorrelated* if, for all t_1 and t_2, we have

$$C_{XY}(t_1, t_2) = 0 \quad \text{or} \quad R_{XY}(t_1, t_2) = \mu_X(t_1)\mu_Y(t_2).$$

They are called *orthogonal* if for all t_1 and t_2,

$$R_{XY}(t_1, t_2) = 0.$$

The Moments of the Stochastic Processes

Theorem

If two stochastic processes are independent, then they are uncorrelated.

Proof. We prove the continuous case here.

$$
\begin{aligned}
C_{XY}(s,t) &= E\left[\big(X_s - \mu_X(s)\big)\big(Y_t - \mu_Y(t)\big)\right] \\
&= \iint (x - \mu_X(s))(y - \mu_Y(t)) f_{XY}(x, y; s, t)\, dx dy \\
&= \int (x - \mu_X(s)) f_X(x, s) dx \cdot \int (y - \mu_Y(t)) f_Y(y, t) dy \\
&= E\big(X_s - \mu_X(s)\big) E\big(Y_t - \mu_Y(t)\big) = 0.
\end{aligned}
$$

$\square$

The Moments of the Stochastic Processes

Example

The processes $\{X_t\}$ and $\{Y_t\}$ are given by

$$X_t = U\cos t + V\sin t,\ t \in (-\infty, +\infty),$$
$$Y_t = U\sin t + V\cos t,\ t \in (-\infty, +\infty),$$

where U and V are two independent random variables, with $E(U) = E(V) = 0$, $E(U^2) = E(V^2) = \sigma^2$. Find the cross-correlation function $R_{XY}(s,t)$.

Solution. The cross-correlation function $R_{XY}(s,t)$ is

$$\begin{aligned} R_{XY}(s,t) &= E[(U\cos s + V\sin s)(U\sin t + V\cos t)] \\ &= E(U^2)\cos s\sin t + E(UV)(\cos s\cos t + \sin s\sin t) + E(V^2) \\ &= \sigma^2\sin(s+t). \end{aligned}$$

□

The Moments of the Stochastic Processes

Example

Suppose that X_t is a signaling process and Y_t is a noise process. Let $W_t = X_t + Y_t$. Find the mean and autocorrelation of W_t.

Solution. The mean function of W_t is

$$\mu_W(t) = \mu_X(t) + \mu_Y(t),$$

and the autocorrelation function is

$$\begin{aligned} R_W(s,t) &= E[(X_s + Y_s)(X_t + Y_t)] \\ &= R_X(s,t) + R_{XY}(s,t) + R_{YX}(s,t) + R_Y(s,t). \end{aligned}$$

In particular, if the mean functions of X_t and Y_t are 0, then they are uncorrelated, and hence

$$R_W(s,t) = R_X(s,t) + R_Y(s,t).$$

$\square$

Contents

1. Definition and Classification
2. The Distribution Family and the Moment Functions
3. The Moments of the Stochastic Processes
4. Stochastic Analysis

Stochastic Analysis

In fact, in many situations, we want to use stochastic processes to model dynamical systems, where we know that the dynamics are continuous in time (i.e., the index set is $\mathbb{R}$, or maybe $\mathbb{R}^+$ or $[0, T]$ for some real T). This means that we ought to restrict the sample paths to be continuous functions; in some cases we would even want them to be differentiable.

To define continuity, it is desirable to have a measure of dissimilarity or "distance" between random variables X and Y. Here we choose the mean of the squared difference, $E\left[(X-Y)^2\right]$, or what is even better, since it resumes the original units of measurement, the square root of this mean square difference. To enhance the suggestion of distance, we introduce the notation

$$\|X\| = \sqrt{E(X^2)} \quad \text{and} \quad \|X-Y\| = \sqrt{E\left[(X-Y)^2\right]}.$$

For the present discussion we will assume that all processes are bounded in Hilbert space, that is, $E[X_t^2] < \infty$. Here we mainly concern continuity in mean.

Definition

(Continuity in Mean) A stochastic process X_t is ***continuous in mean square*** *at t_0 if*

$$\lim_{\varepsilon \to 0} E\left[(X_{t_0+\varepsilon} - X_{t_0})^2\right] = 0$$

Denote it by

$$\text{l.i.m.}_{t \to t_0} X_t = X_{t_0} \quad \text{or} \quad \|X_t - X_{t_0}\| \xrightarrow{t \to t_0} 0,$$

where l.i.m. *denotes limit in mean. The stochastic process $\{X_t\}$ is called a mean square continuous process if it is continuous in mean square for all t.*

Theorem

If X_t is a mean square continuous process, then its mean is also continuous (but not mean square continuous), i.e.,

$$\lim_{\varepsilon \to 0} \mu_X(t+\varepsilon) = \mu_X(t),$$

which can be written as

$$\lim_{\varepsilon \to 0} E(X_{t+\varepsilon}) = E\left(\lim_{\varepsilon \to 0} X_{t+\varepsilon}\right).$$

Proof. By using Cauchy-Schwartz inequality,

$$\begin{aligned} |\mu_X(t+\varepsilon) - \mu_X(t)| &= \left|E\big(X_{t+\varepsilon} - X_t\big)\right| \\ &\leqslant \sqrt{E\big(X_{t+\varepsilon} - X_t\big)^2} \to 0 \quad \text{as } \varepsilon \to 0. \end{aligned}$$

$\square$

Theorem

A stochastic process X_t is a mean square continuous process if and only if its autocorrelation function $R_X(t_1, t_2)$ is continuous.

Proof. By simple calculation, we have

$$\begin{aligned} E\Big(\big(X_{t+\varepsilon} - X_t\big)^2\Big) &= E\left(X_{t+\varepsilon}^2 - 2X_{t+\varepsilon}X_t + X_t^2\right) \\ &= R_X(t+\varepsilon, t+\varepsilon) - 2R_X(t+\varepsilon, t) + R_X(t,t). \end{aligned}$$

If the autocorrelation function $R_X(t_1, t_2)$ is continuous, then X_t is continuous. If X_t is continuous, then we get

$$\text{l.i.m.}_{\varepsilon\to 0} X_{t+\varepsilon} = X_t, \quad \text{l.i.m.}_{h\to 0} X_{t+h} = X_t.$$

By using Cauchy-Schwarz inequality,

$$
\begin{aligned}
& |R_X(t+\varepsilon, t+h) - R_X(t,t)| \\
= \; & |E\left(X_{t+\varepsilon} X_{t+h}\right) - E(X_t X_t)| \\
= \; & |E\left(X_{t+\varepsilon} X_{t+h}\right) - E\left(X_{t+\varepsilon} X_t\right) + E\left(X_{t+\varepsilon} X_t\right) - E(X_t X_t)| \\
\leqslant \; & |E\left(X_{t+\varepsilon} X_{t+h}\right) - E\left(X_{t+\varepsilon} X_t\right)| + |E\left(X_{t+\varepsilon} X_t\right) - E(X_t X_t)| \\
= \; & \left|E\left(X_{t+\varepsilon}\left(X_{t+h} - X_t\right)\right)\right| + \left|E\left(X_t\left(X_{t+\varepsilon} - X_t\right)\right)\right| \\
= \; & \sqrt{E\left(X_{t+\varepsilon}^2\right)}\sqrt{E\left(\left(X_{t+h} - X_t\right)^2\right)} + \sqrt{E\left(X_t^2\right)}\sqrt{E\left(\left(X_{t+\varepsilon} - X_t\right)^2\right)} \\
& \longrightarrow 0 \qquad \text{as } \varepsilon, h \to 0,
\end{aligned}
$$

which means the autocorrelation function $R_X(t_1, t_2)$ of X_t is continuous. $\square$

Definition

Suppose that X_t is a mean square continuous process. We call X_t has a ***mean square derivative*** $X_t^{'}$ *if*

$$\text{l.i.m.}_{\varepsilon \to 0} \frac{X_{t+\varepsilon} - X_t}{\varepsilon} = X_t^{'},$$

that is

$$\lim_{\varepsilon \to 0} E\left(\left(\frac{X_{t+\varepsilon} - X_t}{\varepsilon} - X_t^{'}\right)^2\right) = 0.$$

Theorem

Suppose that the mean and the autocorrelation function of X_t are $\mu_X(t)$ and $R_X(s,t)$ respectively. Then

$$\mu_{X'}(t) = E\left(X^{'}(t)\right) = \frac{d}{dt}E(X_t) = \mu^{'}_X(t),$$

and

$$R_{X'}(s,t) = E\big(X^{'}_s X^{'}_t\big) = \frac{\partial^2 R_X(s,t)}{\partial s \partial t}.$$

Stochastic Analysis

Definition

A mean square integral of a random process X_t is defined by

$$Y_t = \int_{t_0}^{t} X_s ds = \text{l.i.m.}_{\triangle s_i \to 0} \sum_i X(s_i) \triangle s_i,$$

where $t_0 < s_1 < \cdots < t$ and $\triangle s_i = s_{i+1} - s_i$.

Theorem

Suppose that the mean and the autocorrelation function of

$$Y_t = \int_{t_0}^{t} X_s ds$$

are $\mu_Y(t)$ *and* $R_Y(t_1, t_2)$ *respectively. Then*

$$\mu_Y(t) = \int_{t_0}^{t} \mu_X(s) ds,$$

and

$$R_Y(t_1, t_2) = \int_{t_0}^{t_1} \int_{t_0}^{t_2} R_X(s_1, s_2) ds_1 ds_2.$$

Generally, we have

Definition

Suppose that X_t is a stochastic process and $f(t)$ is a real-valued continuous function. The mean square integral of $f(t)X_t$ is defined by

$$Y_t = \int_{t_0}^{t} f(s)X_s ds = \text{l.i.m.}_{|\triangle|\to 0} \sum_i f(s_i)X(s_i)\triangle s_i,$$

where $t_0 < s_1 < \cdots < s_n = t$, $\triangle s_i = s_{i+1} - s_i$ and $|\triangle| = \max\{\triangle s_i, i = 1, 2, \cdots n-1\}$.

Theorem

Suppose that X_t is a mean square continuous process, and $f(t)$ and $g(t)$ are real-valued continuous functions. Then

$$E\left(\int_{t_0}^{t} f(s)X_s ds\right) = \int_{t_0}^{t} f(s)E\big(X_s\big)ds,$$

and

$$E\left(\int_{t_0}^{t_1} f(s_1)X(s_1)ds_1 \int_{t_0}^{t_2} g(s_2)X(s_2)ds_2\right)$$
$$= \int_{t_0}^{t_1}\int_{t_0}^{t_2} f(s_1)g(s_2)E\big(X_{s_1}X_{s_2}\big)ds_1 ds_2.$$

The end

Thank you !